GHENT UNIVERSITY

FACULTY OF BIOSCIENCE ENGINEERING

Exercises Hydrological Modelling

Academic year: 2025 - 2026

Olivier Bonte Hans Lievens

2026-05-26

Table of contents

1 Introduction 3

2 Practical information 4
- 2.1 Working with Quarto 4
- 2.2 Evaluation 4

3 Study area and data 5

4 Exercise 1: Delineation of the Zwalm catchment 6

5 Exercise 2: Implementation of the model 7

6 Exercise 3: Calibration of the model 8

7 Exercise 4:Uncertainty analysis of the model output 9

8 Exercise 5: Data assimilation 10

References 11

1 Introduction

This book contains exercises for the course Hydrological Modelling at Ghent University. The focus of this set of exercises will be on rainfall-runoff modelling with a conceptual hydrological model.

The objective of the practical exercises of this course is to illustrate some of the most important steps necessary to perform a proper modelling of a medium-scale catchment in Flanders. Only some specific steps of the process will be illustrated here across 5 exercises:

1. Delineation of the catchment (Chapter 4)
2. Implementation of a hydrological model (Chapter 5)
3. Calibration of the hydrological model (Chapter 6)
4. The analysis of uncertainty in the model output of the model (Chapter 7)
5. Assimilation of observations into the model (Chapter 8).

Needless to say that for each step, there are many options, and that the choice for e.g. a specific model, calibration algorithm, or evaluation criterium depends on the specific situation, the aim of the modelling exercise, and the personal taste of the modeller. For an extensive – yet non-exhaustive – overview of different techniques in hydrological modelling, we refer to the theoretical course notes and specialized literature.

The aim of the practicals is to:

1. Make you acquainted with the advantages and disadvantages of the selected algorithms,
2. Make you aware of the potential problems related to modelling rainfall-runoff relationships, and
3. Point you to and make you reflect on existing alternatives for the algorithms selected here.

2 Practical information

2.1 Working with Quarto

2.2 Evaluation

3 Study area and data

For these exercises, a hydrological dataset collected in the Zwalm catchment will be used. The Zwalm catchment is located in the center of Belgium and can be considered as a medium-scale catchment.

4 Exercise 1: Delineation of the Zwalm catchment

5 Exercise 2: Implementation of the model

6 Exercise 3: Calibration of the model

7 Exercise 4:Uncertainty analysis of the model output

8 Exercise 5: Data assimilation

References